Matrices & Vectors

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Addition and scalar multiplication**

Let A be the matrix with rows $\langle 1, 3\rangle$ and $\langle -2, 4\rangle$, and B be the matrix with rows $\langle 2, -1\rangle$ and $\langle 0, 5\rangle$. Find $A + B$ and $2A$.

$A + B$ has rows $\langle 3, 2\rangle$ and $\langle -2, 9\rangle$. $2A$ has rows $\langle 2, 6\rangle$ and $\langle -4, 8\rangle$.

2 **[Foundational]** • **Target: Matrix multiplication**

Using A (rows $\langle 1, 3\rangle$, $\langle -2, 4\rangle$) and B (rows $\langle 2, -1\rangle$, $\langle 0, 5\rangle$) from above, find AB.

AB has rows $\langle 2, 14\rangle$ and $\langle -4, 22\rangle$: e.g. row 1, column 1 is $(1)(2) + (3)(0) = 2$.

3 **[Foundational]** • **Target: Matrix multiplication**

Let A have rows $\langle 1, 3\rangle$ and $\langle -2, 4\rangle$, and let $\mathbf{v} = \langle 3, -1\rangle$. Find $A\mathbf{v}$.

$A\mathbf{v} = \langle (1)(3) + (3)(-1),\ (-2)(3) + (4)(-1)\rangle = \langle 0, -10\rangle$.

4 **[Foundational]** • **Target: Determinant and inverse**

Find $\det A$ for the matrix A with rows $\langle 1, 3\rangle$ and $\langle -2, 4\rangle$, and state whether A is invertible.

$\det A = (1)(4) - (3)(-2) = 4 + 6 = 10 \neq 0$, so A is invertible.

5 **[Foundational]** • **Target: Determinant and inverse**

Find A^{-1} for the matrix A with rows $\langle 1, 3\rangle$ and $\langle -2, 4\rangle$.

$A^{-1} = \frac{1}{10} \times$ the matrix with rows $\langle 4, -3\rangle$ and $\langle 2, 1\rangle$, i.e., A^{-1} has rows $\langle 0.4, -0.3\rangle$ and $\langle 0.2, 0.1\rangle$.

6 **[Foundational]** • **Target: Solving systems with the inverse matrix**

Solve the system $x + 3y = 13$ and $-2x + 4y = 4$ using $\mathbf{x} = A^{-1}\mathbf{b}$ (using A^{-1} from above).

A^{-1} has rows $\langle 0.4, -0.3\rangle$ and $\langle 0.2, 0.1\rangle$. Then $x = 0.4(13) - 0.3(4) = 5.2 - 1.2 = 4$ and $y = 0.2(13) + 0.1(4) = 2.6 + 0.4 = 3$. Check: $4 + 3(3) = 13$ and $-2(4) + 4(3) = 4$.

7 **[Foundational]** • **Target: Angle between vectors**

Find the angle between $\mathbf{p} = \langle 2, 2\rangle$ and $\mathbf{q} = \langle 3, 0\rangle$.

$\mathbf{p} \cdot \mathbf{q} = 6$, $|\mathbf{p}| = 2\sqrt{2}$, $|\mathbf{q}| = 3$. $\cos\theta = \frac{6}{6\sqrt{2}} = \frac{1}{\sqrt{2}} \approx 0.7071 \Rightarrow \theta = 45^\circ$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Angle between vectors**

Find the angle between $\langle 7, 1\rangle$ and $\langle 2, -3\rangle$, rounded to the nearest tenth of a degree.

Dot product $= 14 - 3 = 11$, magnitudes $\sqrt{50}$ and $\sqrt{13}$. $\cos\theta = \frac{11}{\sqrt{650}} \approx 0.4314 \Rightarrow \theta \approx 64.4^\circ$.

9 **[Intermediate]** • **Target: Determinant and inverse**

Given 3D vectors $\mathbf{u} = \langle 2, -3, 6\rangle$ and $\mathbf{v} = \langle 1, 4, 2\rangle$: (a) Find the unit vector in the direction of $\mathbf{u}$. (b) Calculate the angle θ between $\mathbf{u}$ and $\mathbf{v}$ to the nearest degree.

Part (a): Find the unit vector $\hat{\mathbf{u}}$: 1. Magnitude of $\mathbf{u}$: $|\mathbf{u}| = \sqrt{2^2 + (-3)^2 + 6^2} = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$. 2. Unit vector: $\hat{\mathbf{u}} = \frac{\mathbf{u}}{|\mathbf{u}|} = \left\langle \frac{2}{7}, -\frac{3}{7}, \frac{6}{7}\right\rangle$. Part (b): Find the angle θ: 1. Dot product $\mathbf{u} \cdot \mathbf{v}$: $\mathbf{u} \cdot \mathbf{v} = (2)(1) + (-3)(4) + (6)(2) = 2 - 12 + 12 = 2$. 2. Magnitude of $\mathbf{v}$.

10 **[Intermediate]** • **Target: Solving systems with the inverse matrix**

Given vectors $\mathbf{a} = 2\mathbf{i} - 5\mathbf{j}$ and $\mathbf{b} = -3\mathbf{i} + 4\mathbf{j}$, find the scalar projection of $\mathbf{a}$ onto $\mathbf{b}$ and the angle θ between them to the nearest tenth of a degree.

Calculate the dot product: $\mathbf{a} \cdot \mathbf{b} = (2)(-3) + (-5)(4) = -6 - 20 = -26$. Calculate the magnitudes: $|\mathbf{a}| = \sqrt{2^2 + (-5)^2} = \sqrt{4 + 25} = \sqrt{29} \approx 5.385$, $|\mathbf{b}| = \sqrt{(-3)^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. Scalar projection of $\mathbf{a}$ onto $\mathbf{b}$: $\text{comp}_{\mathbf{b}}\mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|} = \frac{-26}{5} = -5.2$. Angle θ between vectors.

11 **[Intermediate]** • **Target: Vector magnitude and dot product**

Find all real values of the constant k for which the matrix $M = (k + 2, 4; 3, k - 2)$ is singular (has no inverse).

Step 1: Set the determinant to zero: A matrix is singular if and only if $\det(M) = 0$: $\det(M) = (k + 2)(k - 2) - (4)(3) = 0$. Step 2: Solve the algebraic equation: $(k^2 - 4) - 12 = 0 \Rightarrow k^2 - 16 = 0$. Step 3: Extract roots: $(k - 4)(k + 4) = 0 \Rightarrow \mathbf{k = 4}$ or $\mathbf{k = -4}$. Check: For $k = 4$, $M = (6, 4; 3, 2)$, $\det = 12 - 12 = 0$ ✓. For $k = -4$, $M = (-2, 4; 3, -6)$, $\det = 12 - 12 = 0$ ✓.

12 **[Intermediate]** • **Target: Angle between vectors**

Find the angle θ between the vectors $\mathbf{u} = \langle 1, \sqrt{3}\rangle$ and $\mathbf{v} = \langle 2, 0\rangle$ in degrees.

Step 1: Compute dot product and magnitudes: $\mathbf{u} \cdot \mathbf{v} = (1)(2) + (\sqrt{3})(0) = 2$. $||\mathbf{u}|| = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{1 + 3} = 2$, $||\mathbf{v}|| = \sqrt{2^2 + 0^2} = 2$. Step 2: Solve for $\cos\theta$: $\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}||||\mathbf{v}||} = \frac{2}{(2)(2)} = \frac{2}{4} = \frac{1}{2} \Rightarrow \theta = \arccos(1/2) = 60^\circ$.

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13 [Intermediate] • Target: Addition and scalar multiplication

Find the magnitude and direction angle θ (in degrees, $0^\circ \le \theta < 360^\circ$) of the vector $\mathbf{v} = \langle -5, 5\sqrt{3} \rangle$.

Step 1: Compute magnitude: $|\mathbf{v}| = \sqrt{(-5)^2 + (5\sqrt{3})^2} = \sqrt{25 + 75} = \sqrt{100} = \mathbf{10}$. Step 2: Direction angle: Since $v_x = -5 < 0$ and $v_y = 5\sqrt{3} > 0$, the vector lies in Quadrant II. Reference angle: $\alpha = \tan^{-1}\left(\frac{5\sqrt{3}}{5}\right) = \tan^{-1}(\sqrt{3}) = 60^\circ$. $\theta = 180^\circ - 60^\circ = \mathbf{120}^\circ$.

14 [Intermediate] • Target: Matrix multiplication

Find the angle θ between the vectors $\mathbf{a} = \langle 1, 2, 2 \rangle$ and $\mathbf{b} = \langle 2, -2, 1 \rangle$.

Step 1: Compute dot product: $\mathbf{a} \cdot \mathbf{b} = (1)(2) + (2)(-2) + (2)(1) = 2 - 4 + 2 = 0$. Step 2: Conclude angle: Since $\mathbf{a} \cdot \mathbf{b} = 0$ and both vectors have non-zero magnitude ($|\mathbf{a}| = 3$, $|\mathbf{b}| = 3$): $\cos\theta = 0 \Rightarrow \theta = \mathbf{90}^\circ$ $\left(\text{or } \frac{\pi}{2} \text{ rad}\right)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Determinant and inverse • Cross-Lesson: 3.01 Functions & Graphs

Find the inverse of the matrix $A = (4, 7; 1, 2)$.

Step 1: Compute determinant: $\det(A) = (4)(2) - (7)(1) = 8 - 7 = 1$. Step 2: Apply the inverse formula: $A^{-1} = \frac{1}{\det(A)}(d, -b; -c, a) = \frac{1}{1}(2, -7; -1, 4) = (\mathbf{2, -7; -1, 4})$. Check: $AA^{-1} = (8 - 7, -28 + 28; 2 - 2, -7 + 8) = (1, 0; 0, 1) = I$ ✓.

16 [Challenging] • Target: Solving systems with the inverse matrix • Cross-Lesson: 5.01 Introductory Calculus

Compute the matrix product AB where $A = (1, 3; -2, 4)$ and $B = (2, 0; 5, -1)$.

Step 1: Compute rows of A dot columns of B: $(AB)_{11} = (1)(2) + (3)(5) = 2 + 15 = 17$ $(AB)_{12} = (1)(0) + (3)(-1) = 0 - 3 = -3$ $(AB)_{21} = (-2)(2) + (4)(5) = -4 + 20 = 16$ $(AB)_{22} = (-2)(0) + (4)(-1) = 0 - 4 = -4$ Conclusion: $AB = (\mathbf{17, -3; 16, -4})$.

17 [Challenging] • Target: Vector magnitude and dot product • Cross-Lesson: 6.01 Probability

Rotate the Cartesian axes to eliminate the cross-product term in $5x^2 - 4xy + 8y^2 = 36$. Find the angle of rotation θ, standard form in the rotated system (x', y'), and the lengths of the semi-major and semi-minor axes.

Step 1: Determine the rotation angle θ: Here $A = 5$, $B = -4$, and $C = 8$. $\cot(2\theta) = \frac{A - C}{B} = \frac{5 - 8}{-4} = \frac{-3}{-4} = \frac{3}{4} > 0$. Since $\cot(2\theta) = 3/4$, by right triangle trigonometry $\cos(2\theta) = \frac{3}{5}$. Using half-angle formulas for $\theta \in (0, \pi/4)$: $\cos\theta = \sqrt{\frac{1 + \cos(2\theta)}{2}} = \sqrt{\frac{1 + 3/5}{2}} = \sqrt{\frac{4}{5}} = \frac{2}{\sqrt{5}}$,.

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18 [Challenging] • Target: Angle between vectors • Cross-Lesson: 1.01 Complex Numbers

Vectors $\mathbf{a} = \langle 4, k \rangle$ and $\mathbf{b} = \langle 6, -8 \rangle$ are perpendicular. Find the value of k, and compute the projection of $\mathbf{c} = \langle 10, 5 \rangle$ onto $\mathbf{b}$.

Step 1: Use the orthogonality condition $\mathbf{a} \cdot \mathbf{b} = 0$: $\mathbf{a} \cdot \mathbf{b} = (4)(6) + (k)(-8) = 24 - 8k = 0 \Rightarrow 8k = 24 \Rightarrow \mathbf{k = 3}$. Step 2: Compute the vector projection of $\mathbf{c}$ onto $\mathbf{b}$: The scalar projection is $\text{comp}_{\mathbf{b}}\mathbf{c} = \frac{\mathbf{c} \cdot \mathbf{b}}{|\mathbf{b}|}$. 1. $\mathbf{c} \cdot \mathbf{b} = (10)(6) + (5)(-8) = 60 - 40 = 20$. 2. $|\mathbf{b}| = \sqrt{6^2 + (-8)^2} = \sqrt{36 + 64} = \sqrt{100} = 10$. 3. Vec.

19 [Challenging] • Target: Addition and scalar multiplication • Cross-Lesson: 2.01 Sequences & Series

Let $A = (3, 1; 5, 2)$. Find A^{-1} and use it to solve the linear system: $3x + y = 7$ $5x + 2y = 12$

Step 1: Compute the determinant of A: $\det(A) = ad - bc = (3)(2) - (1)(5) = 6 - 5 = 1$. Step 2: Construct the inverse A^{-1}: Swap the main diagonal entries, negate the off-diagonal entries, and divide by $\det(A) = 1$: $A^{-1} = \frac{1}{1}(2, -1; -5, 3) = (\mathbf{2, -1; -5, 3})$. Step 3: Solve the matrix equation $A\mathbf{x} = \mathbf{b}$.

20 [Challenging] • Target: Matrix multiplication • Cross-Lesson: 3.01 Functions & Graphs

Calculate the exact value of $\cos(75^\circ)$ using angle addition, and evaluate $\tan(2\theta)$ given that $\tan\theta = \frac{2}{3}$.

Part 1: Exact value of $\cos(75^\circ)$: Express $75^\circ = 45^\circ + 30^\circ$ and apply the cosine sum identity: $\cos(45^\circ + 30^\circ) = \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ = \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right) = \frac{\sqrt{6} - \sqrt{2}}{4} = \frac{\mathbf{\sqrt{6} - \sqrt{2}}}{\mathbf{4}}$. Part 2: Evaluate $\tan(2\theta)$: Apply the double-angle identity for tangent.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Addition and scalar multiplication

Consider a linear transformation T represented by the matrix $A = (2, 1\ 1, 3)$. Let $P = (x; y)$ be a point in the plane. The transformation maps P to P'. If the area of the region transformed by T from a unit square in the xy-plane is A, and the determinant of A is $\det(A)$, find the value of $\det(A)$ if the area of the transformed region is 5.

a. **5** ✓

b. 2

c. 1

d. 10

The area of the region transformed by a linear transformation T represented by a matrix A from a unit square is given by the absolute value of the determinant of A, i.e., Area $= |\det(A)|$. In this problem, we are given that the matrix representing the transformation is $A = (2, 1\ 1, 3)$. The determinant of A is calculated as: $\det(A) = (2 \times 3) - (1 \times 1) = 6 - 1 = 5$. We are also given that the area of the transformed region is 5. Thus, $|\det(A)| = 5$. Since we calculated.

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22 [Foundational] • Target: Matrix multiplication

Consider the matrix $A = (2, 0; 0, 3)$. When the matrix A is applied to the vector $\mathbf{v} = (x; y)$, the resulting vector is $A\mathbf{v} = (6; 9)$. What is the original vector $\mathbf{v}$?

a. $(3; 3)$ ✓

b. $(1; 2)$

c. $(6; 3)$

d. $(12; 27)$

The matrix multiplication is given by: $(2, 0; 0, 3)(x; y) = (2x; 3y)$ We are given that this result is $(6; 9)$. Therefore, we have the system of equations: $2x = 6$ $3y = 9$ Solving for x and y: $x = \frac{6}{2} = 3$ $y = \frac{9}{3} = 3$ So, the original vector $\mathbf{v}$ is $(3; 3)$.

23 [Foundational] • Target: Determinant and inverse

Find the angle θ between $\mathbf{u} = \langle 1, 7 \rangle$ and $\mathbf{v} = \langle -7, 4 \rangle$, to the nearest 0.1°.

a. 68.4° ✓

b. 111.6°

c. 78.4°

d. 63.4°

$\mathbf{u} \cdot \mathbf{v} = 21$, $|\mathbf{u}| = \sqrt{50}$, $|\mathbf{v}| = \sqrt{65}$. $\cos\theta = \frac{21}{\sqrt{50}\sqrt{65}} \approx 0.3684$, so $\theta \approx 68.4^\circ$ (Theorem 2.2).

24 [Foundational] • Target: Solving systems with the inverse matrix

Solve using the inverse-matrix method: $\{-4x + 5y = 17; 2x + 2y = -4$.

a. **x=-3,\ y=1** ✓

b. x=1,\ y=-3

c. x=-2,\ y=1

d. x=-3,\ y=-1

$\det A = -18$. Using A^{-1} (Theorem 2.1) on $\mathbf{b} = [17;\ -4]$ gives $x = -3,\ y = 1$. Check: $-4(-3) + 5(1) = 17$ ✓ and $2(-3) + 2(1) = -4$ ✓.

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25 [Foundational] • Target: Vector magnitude and dot product

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in the coordinate plane such that $|\mathbf{u}| = 5$, $|\mathbf{v}| = 10$, and the angle θ between them is obtuse ($90^\circ < \theta < 180^\circ$). If the minimum value of $|t\mathbf{u} + \mathbf{v}|$ over all real numbers t is equal to 6, what is the value of the dot product $\mathbf{u} \cdot \mathbf{v}$?

a. -50

b. **-40 ✓**

c. -30

d. 40

We can solve this problem algebraically or geometrically. **Method 1: Algebraic (Quadratic Optimization)** Consider the square of the magnitude $|t\mathbf{u} + \mathbf{v}|$: $|t\mathbf{u} + \mathbf{v}|^2 = (t\mathbf{u} + \mathbf{v}) \cdot (t\mathbf{u} + \mathbf{v}) = t^2|\mathbf{u}|^2 + 2t(\mathbf{u} \cdot \mathbf{v}) + |\mathbf{v}|^2$ Let $D = \mathbf{u} \cdot \mathbf{v}$. Substituting $|\mathbf{u}| = 5$ and $|\mathbf{v}| = 10$: $f(t) = 25t^2 + 2Dt + 100$ This is a quadratic function in t with leading coefficien.

26 [Foundational] • Target: Angle between vectors

A matrix $M = (k, 2; 3, 1)$ transforms a point $P(1, 2)$ to $P'(5, 5)$. Find the value of k.

a. 1

b. **2 ✓**

c. 3

d. 4

The transformation of a point $P(x, y)$ by a matrix $M = (a, b; c, d)$ is given by MP. In this case, we have $(k, 2; 3, 1)(1; 2) = (5; 5)$. Performing the matrix multiplication, we get: $k(1) + 2(2) = 5$ and $3(1) + 1(2) = 5$. The second equation, $3 + 2 = 5$, is consistent. For the first equation, we have $k + 4 = 5$. Solving for k, we subtract 4 from both sides: $k = 5 - 4$, which gives $k = 1$. Therefore, the value of k is 1.

27 [Foundational] • Target: Addition and scalar multiplication

A vector $\mathbf{v}$ is represented by the directed line segment from point $P(1, 2)$ to point $Q(4, 6)$. What are the components of vector $\mathbf{v}$?

a. **[3, 4] ✓**

b. [5, 8]

c. [-3, -4]

d. [4, 3]

The components of a vector $\mathbf{v}$ represented by a directed line segment from point $P(x_1, y_1)$ to point $Q(x_2, y_2)$ are given by $\mathbf{v} = [x_2 - x_1, y_2 - y_1]$. In this case, $P = (1, 2)$ and $Q = (4, 6)$. Therefore, the components of $\mathbf{v}$ are $[4 - 1, 6 - 2] = [3, 4]$.

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28 [Foundational] • Target: Matrix multiplication

In the standard (x, y)-coordinate plane, vectors $\mathbf{u} = \langle a, b \rangle$ and $\mathbf{v} = \langle c, d \rangle$ have magnitudes $||\mathbf{u}|| = 13$ and $||\mathbf{v}|| = 5$. If their dot product is $\mathbf{u} \cdot \mathbf{v} = 25$, and the matrix $M = [a, b; c, d]$ has $\det(M) > 0$, what is the value of $\det(M)$?

a. 25

b. 40

c. **60** ✓

d. 65

We can solve this using either Lagrange's algebraic identity or the geometric interpretation of the dot product and determinant. **Method 1: Lagrange's Identity** For any real numbers a, b, c, d, the identity connecting the determinant and the dot product is: $(ad - bc)^2 + (ac + bd)^2 = (a^2 + b^2)(c^2 + d^2)$ Notice that: - $\det(M) = ad - bc$ - $\mathbf{u} \cdot \mathbf{v} = ac + bd = 25$ - $||\mathbf{u}||^2 = a^2 + b^2 = 13^2 = 169$ - $||\mathbf{v}||^2 = c^2 + d^2 = 5^2 = 25$ Substituting these values into th.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Determinant and inverse

For all real values of θ, the matrix product $[\cos\theta, \sin\theta][6, 2; 2, 3][\cos\theta; \sin\theta]$ evaluates to a 1×1 matrix $[k]$. What is the maximum possible value of k?

a. 6

b. **7** ✓

c. 9

d. 14

Step 1: Perform the matrix multiplication. First, multiply the 1×2 row vector by the 2×2 matrix: $[\cos\theta, \sin\theta][6, 2; 2, 3] = [6\cos\theta + 2\sin\theta, 2\cos\theta + 3\sin\theta]$ Next, multiply this result by the 2×1 column vector: $k = (6\cos\theta + 2\sin\theta)\cos\theta + (2\cos\theta + 3\sin\theta)\sin\theta$ $k = 6\cos^2\theta + 4\sin\theta\cos\theta + 3\sin^2\theta$ Step 2: Express the quadratic trigonometric terms using double-angle.

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30 [Intermediate] • Target: Solving systems with the inverse matrix

Matrices A and B are defined as $A = [2, k; -1, 3]$ and $B = [1, 4; 2, k]$, where k is a real number. If the determinant of the matrix sum $A + B$ is equal to 17, what is the value of k? (Note: The determinant of a 2×2 matrix $[a, b; c, d]$ is defined as $ad - bc$.)

a. 1

b. 2

c. **6** ✓

d. $\frac{19}{2}$

Step 1: Compute the matrix sum $A + B$ by adding corresponding entries:

$A + B = [2 + 1, k + 4; -1 + 2, 3 + k] = [3, k + 4; 1, k + 3]$ Step 2: Calculate the determinant of $A + B$ using the definition $\det[a, b; c, d] = ad - bc$: $\det(A + B) = (3)(k + 3) - (1)(k + 4)$ Distribute carefully across both terms: $\det(A + B) = 3k + 9 - k - 4 = 2k + 5$ Step 3: Set the determinant equal to 17 and solve for k: $2k + 5 = 17 \Rightarrow 2k = 12 \Rightarrow k = 6$ Common Misconception.

31 [Intermediate] • Target: Vector magnitude and dot product

For a 2×2 matrix $[a, b; c, d]$, the determinant is defined as $ad - bc$. Given the matrices $A = [x, 2; -1, 3]$ and $B = [1, -4; 3, x]$, the determinant of the matrix $A + B$ is equal to 19. If $x > 0$, what is the value of x?

a. **2** ✓

b. $\frac{5}{4}$

c. 4

d. 6

First, find the matrix sum $A + B$ by adding corresponding entries:

$A + B = [x + 1, 2 + (-4); -1 + 3, 3 + x] = [x + 1, -2; 2, x + 3]$ Next, calculate the determinant of $A + B$: $\det(A + B) = (x + 1)(x + 3) - (-2)(2)$ $\det(A + B) = (x^2 + 4x + 3) - (-4) = x^2 + 4x + 7$ Set the determinant equal to 19 and solve for x: $x^2 + 4x + 7 = 19$ $x^2 + 4x - 12 = 0$ Factoring the quadratic gives: $(x + 6)(x - 2) = 0$ This yields $x = -6$ or $x = 2$. Since we are given that $x > 0$, the value is.

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32 [Intermediate] • Target: Angle between vectors

In the standard coordinate plane, vector $\mathbf{u} = \langle 2, -1 \rangle$ and vector $\mathbf{v} = \langle k, 5 \rangle$, where k is a real constant. If the magnitude of the vector $\mathbf{u} + \mathbf{v}$ is equal to 5, which of the following is a possible value of k?

a. **-5** ✓

b. -1

c. 3

d. 7

First, find the sum of vectors $\mathbf{u}$ and $\mathbf{v}$ by adding their corresponding components: $\mathbf{u} + \mathbf{v} = \langle 2 + k, -1 + 5 \rangle = \langle 2 + k, 4 \rangle$ Next, use the magnitude formula $||\langle x, y \rangle|| = \sqrt{x^2 + y^2}$: $\sqrt{(2 + k)^2 + 4^2} = 5$ Square both sides of the equation to eliminate the radical: $(2 + k)^2 + 16 = 25$ $(2 + k)^2 = 9$ Take the square root of both sides: $2 + k = 3 \quad \text{or} \quad 2 + k = -3$ $k = 1 \quad \text{or} \quad k = -5$ Amon.

33 [Intermediate] • Target: Addition and scalar multiplication

What is the value of the determinant of the matrix $[7, -3; 2, 4]$?

a. **34** ✓

b. 22

c. -29

d. -34

The determinant of a 2×2 matrix $[a, b; c, d]$ is given by the formula: $\det = ad - bc$ Substitute the entries $a = 7$, $b = -3$, $c = 2$, and $d = 4$: $\det = (7)(4) - (-3)(2) = 28 - (-6) = 28 + 6 = 34$ Common misconceptions: - Calculating $ad + bc$ gives $28 + (-6) = 22$. - Reversing the subtraction order as $bc - ad$ gives $-6 - 28 = -34$. - Multiplying along the rows instead of diagonals gives $(7)(-3) - (2)(4) = -21 - 8 = -29$.

34 [Intermediate] • Target: Matrix multiplication

Given the vectors $\mathbf{u} = \langle 3, -4 \rangle$ and $\mathbf{v} = \langle -2, 5 \rangle$, what is the vector $2\mathbf{u} + \mathbf{v}$?

a. $\langle 4, -3 \rangle$ ✓

b. $\langle 1, 1 \rangle$

c. $\langle 4, 1 \rangle$

d. $\langle 8, -13 \rangle$

To find $2\mathbf{u} + \mathbf{v}$, first perform the scalar multiplication on $\mathbf{u}$: $2\mathbf{u} = 2\langle 3, -4 \rangle = \langle 2(3), 2(-4) \rangle = \langle 6, -8 \rangle$ Next, add the corresponding components of $2\mathbf{u}$ and $\mathbf{v}$: $2\mathbf{u} + \mathbf{v} = \langle 6 + (-2), -8 + 5 \rangle = \langle 4, -3 \rangle$ - $\langle 1, 1 \rangle$ is incorrect because it represents $\mathbf{u} + \mathbf{v}$, omitting the scalar factor of 2. - $\langle 4, 1 \rangle$ is incorrect because the scal.

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35 [Intermediate] • Target: Determinant and inverse

Consider a point $P(\cos\theta, \sin\theta)$ on the unit circle in the Cartesian plane. If the distance from the origin to the point $Q(x, y)$ is 5, and the angle between the positive x-axis and the line segment OQ is α, such that $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. If $x = \cos\alpha$ and $y = \sin\alpha$, find the maximum possible value of the product of the x-coordinates of P and Q, i.e., $\cos\theta \cdot \cos\alpha$.

a. 4/5

b. 3/5

c. **1** ✓

d. 5/3

The point P is on the unit circle, so its coordinates are $(\cos\theta, \sin\theta)$. We are given $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. The point Q has coordinates (x, y) and its distance from the origin is 5. The angle between the positive x-axis and OQ is α. Thus, $x = 5\cos\alpha$ and $y = 5\sin\alpha$. We are asked to find the maximum possible value of the product of the x-coordinates of P and Q, which is $\cos\theta \cdot x$. Substituting the given values, we h.

36 [Intermediate] • Target: Solving systems with the inverse matrix

Consider the vector $\mathbf{v} = \langle 3, 4 \rangle$. What is the magnitude of this vector?

a. **5** ✓

b. 7

c. 12

d. 25

The magnitude of a vector $\mathbf{v} = \langle x, y \rangle$ is given by the formula $|\mathbf{v}| = \sqrt{x^2 + y^2}$. For the given vector $\mathbf{v} = \langle 3, 4 \rangle$, we have x = 3 and y = 4. Substituting these values into the formula: $|\mathbf{v}| = \sqrt{3^2 + 4^2}$ $|\mathbf{v}| = \sqrt{9 + 16}$ $|\mathbf{v}| = \sqrt{25}$ $|\mathbf{v}| = 5$ Therefore, the magnitude of the vector $\mathbf{v}$ is 5.

37 [Intermediate] • Target: Vector magnitude and dot product

If matrix $A = [3, -2; 0, 4]$ and matrix $B = [-1, 5; 2, -3]$, what is the matrix $2A + B$?

a. $[5, 1; 2, 5]$ ✓

b. $[2, 3; 2, 1]$

c. $[5, 3; 2, 1]$

d. $[7, -9; -2, 11]$

To find the matrix $2A + B$, first perform scalar multiplication by multiplying each entry of matrix A by 2: $2A = 2[3, -2; 0, 4] = [2(3), 2(-2); 2(0), 2(4)] = [6, -4; 0, 8]$ Next, add the corresponding entries of $2A$ and B:

$2A + B = [6, -4; 0, 8] + [-1, 5; 2, -3] = [6 + (-1), -4 + 5; 0 + 2, 8 + (-3)] = [5, 1; 2, 5]$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Angle between vectors • Cross-Lesson: 1.01 Complex Numbers

Find the angle between $\mathbf{u} = \langle 2, 1 \rangle$ and $\mathbf{v} = \langle 4, 2 \rangle$ (to one decimal place).

a. **0.0°** ✓

b. 20.0°

c. 180.0°

d. 10.0°

$\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{u}||\mathbf{v}|} = \frac{10}{2.24 \times 4.47} \approx 1.0$, so $\theta \approx 0.0°$.

39 [Challenging] • Target: Addition and scalar multiplication • Cross-Lesson: 2.01 Sequences & Series

A matrix has 1 rows and 3 columns. What are its dimensions?

a. **1×3** ✓

b. 4

c. 3

d. 3×1

Matrix dimensions are written rows×columns: 1×3.

40 [Challenging] • Target: Matrix multiplication • Cross-Lesson: 3.01 Functions & Graphs

A vector $\mathbf{u} = \langle 3, k \rangle$ has magnitude 5. Find the positive value of k, given $k = 4$ is one solution.

a. **4** ✓

b. 2

c. 3

d. 5

$\sqrt{3^2 + k^2} = 5 \Rightarrow k^2 = 16 \Rightarrow k = 4$.

41 [Challenging] • Target: Determinant and inverse • Cross-Lesson: 5.01 Introductory Calculus

Find the determinant of $(-1, 2;\ -6, 4)$.

a. **8** ✓

b. -16

c. 9

d. 7

$\det = ad - bc = (-1)(4) - (2)(-6) = -4 - (-12) = 8$.

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42 [Challenging] • Target: Solving systems with the inverse matrix • Cross-Lesson: 6.01 Probability

A factory produces two types of components, A and B. Components of type A have a defect rate of 3%, and components of type B have a defect rate of 5%. A batch of 100 components is produced, consisting of N components of type A and $100 - N$ components of type B. Let X be the random variable representing the total number of defective components in the batch. If the expected number of defective components is $E[X] = 4.2$, find the probability that a randomly selected component from this batch is defective, given that it is of type A.

a. **0.030** ✓

b. 0.042

c. 0.050

d. 0.036

Let X_A be the number of defective components of type A, and X_B be the number of defective components of type B. The total number of defective components is $X = X_A + X_B$. The expected number of defective components of type A is $E[X_A] = N \times 0.03$. The expected number of defective components of type B is $E[X_B] = (100 - N) \times 0.05$. The total expected number of defective components is $E[X] = E[X_A] + E[X_B]$. We are given $E[X] = 4.2$. So, $N \times 0.03 + (100 - N) \times 0.05 = 4.2$..

43 [Challenging] • Target: Vector magnitude and dot product • Cross-Lesson: 1.01 Complex Numbers

A bag contains 12 red marbles and 8 blue marbles. If one marble is drawn at random, what is the probability that the marble is red?

a. $\frac{3}{5}$ ✓

b. $\frac{2}{5}$

c. $\frac{1}{2}$

d. $\frac{3}{4}$

The total number of marbles in the bag is the sum of red and blue marbles: $12 + 8 = 20$ marbles. The number of favorable outcomes (drawing a red marble) is 12. The probability of an event is calculated as the ratio of the number of favorable outcomes to the total number of possible outcomes. Therefore, the probability of drawing a red marble is $P(\text{red}) = \frac{\text{Number of red marbles}}{\text{Total number of marbles}} = \frac{12}{20}$. This fraction can be simplified by dividing both the nume.

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44 [Challenging] • Target: Angle between vectors • Cross-Lesson: 2.01 Sequences & Series

A fair six-sided die is rolled. What is the probability of rolling a number greater than 4?

a. 1/6

b. **1/3** ✓

c. 1/2

d. 2/3

The possible outcomes when rolling a fair six-sided die are {1, 2, 3, 4, 5, 6}. The total number of possible outcomes is 6. We are interested in the event of rolling a number greater than 4. The numbers in the set of outcomes that are greater than 4 are {5, 6}. There are 2 favorable outcomes. The probability of an event is calculated as the ratio of the number of favorable outcomes to the total number of possible outcomes. Probability (rolling > 4) = (Number of favorable outcomes) / (Total number of possible outcom.

45 [Challenging] • Target: Addition and scalar multiplication • Cross-Lesson: 3.01 Functions & Graphs

A bag contains 5 red marbles and 3 blue marbles. If a marble is drawn at random, what is the probability that it is blue?

a. $\frac{3}{8}$ ✓

b. $\frac{5}{8}$

c. $\frac{3}{5}$

d. $\frac{5}{3}$

The total number of marbles in the bag is the sum of red and blue marbles: $5 + 3 = 8$. The number of favorable outcomes (drawing a blue marble) is 3. The probability of an event is calculated as the ratio of the number of favorable outcomes to the total number of possible outcomes. Therefore, the probability of drawing a blue marble is $\frac{\text{Number of blue marbles}}{\text{Total number of marbles}} = \frac{3}{8}$.

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